

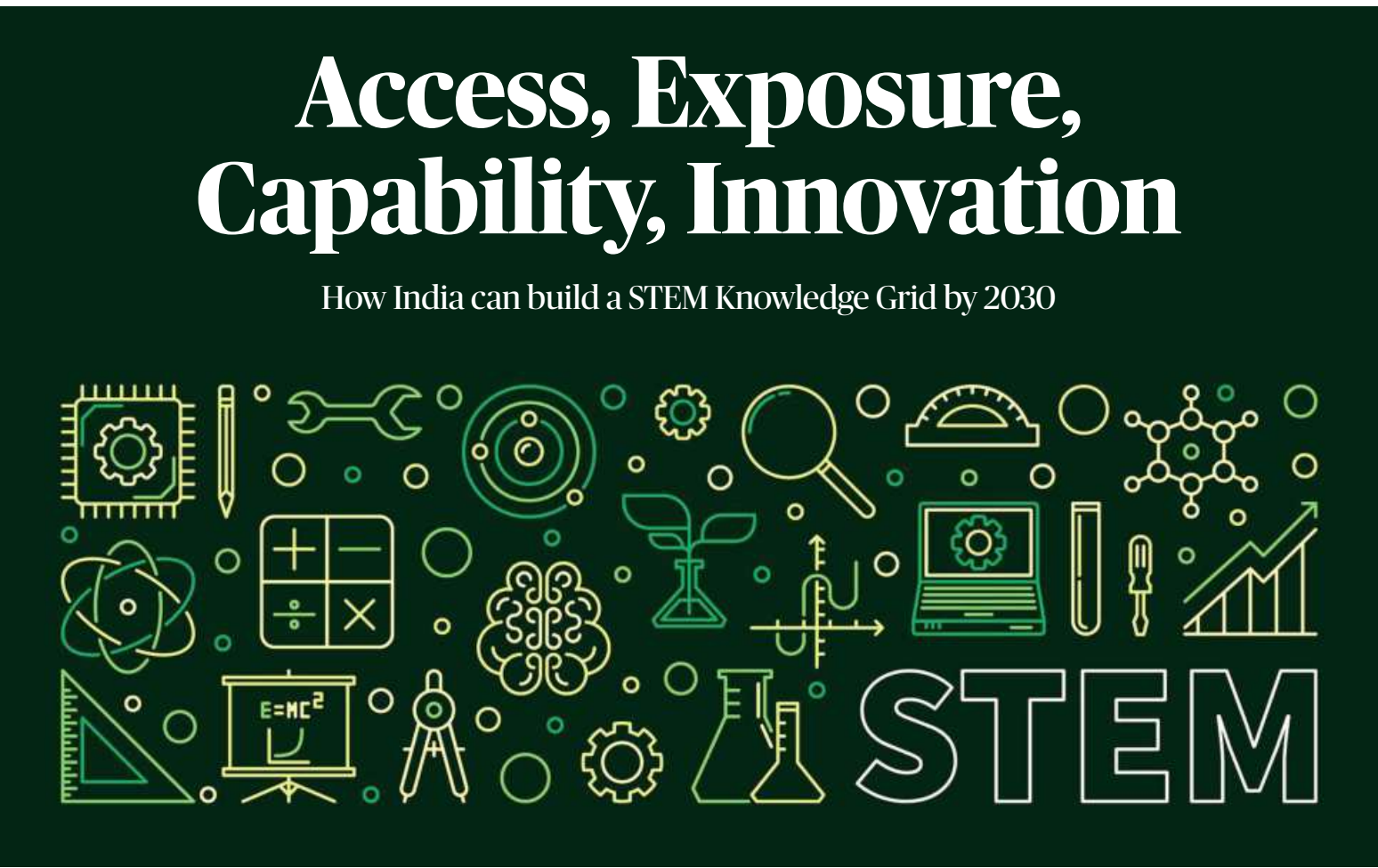
Harilal Bhaskar

India often speaks proudly about its demographic dividend, digital expansion, and deep-tech ambition. Yet beneath the optimism rests a quiet contradiction: we expect 21st-century innovation from an education and research ecosystem still shaped by 20th-century structures.

Many institutions continue to work with ageing laboratories. School science remains theory-centred. Teachers are overburdened. Industry engagement is sporadic. Even our brightest students rarely interact with authentic scientific tools before entering the job market.

If India intends to lead the world in climate tech, biotechnology, advanced manufacturing, defence R&D, and space innovation by 2030, a slow, incremental path will not suffice. We need a decisive shift from the classroom as the primary learning unit to a distributed national knowledge grid, co-created through modern Public-Private Partnerships (PPPs) that are capability building, flexible, modular, and innovation-driven.

Learning Ecosystems: Around the world, the EU's Digital Education Action Plan and America's regional innovation hubs have shown that learning ecosystems – not isolated campuses – build innovation capacity. India needs its own version: Knowledge Grid = Public Infras-



structure + Private Expertise + Student Mobility. This involves institutions providing classrooms, labs, and academic frameworks; industry contributing tools, domain knowledge, mentors, and real problem statements; and students moving seamlessly across these nodes, learning from real challenges rather than outdated theory. Every district becomes a STEM node. Every node connects to national labs and industrial

clusters. Not a new megacampus but a living network.

Industrial Sandbox Rooms: A major barrier in India is late exposure to real scientific tools. Students often encounter equipment for the first time in their twenties, if at all. A rapid reform is to create Industrial Sandbox Rooms in schools and colleges. These are small tactically designed micro-labs equipped with simplified industrial machines, simu-

lators for electronics, chip design, drone systems, and materials, and quarterly "challenge kits" inspired by industry use-cases. Students will learn through experimentation, failure-driven insight, and project-based exploration. Germany's dual-system and South Korea's early-tech exposure models demonstrate the transformative power of such small units. India can adopt the same logic at Indian cost points, with Indian

ingenuity.

Inverted Internships: One-week internships cannot build competence. A more realistic model is the Inverted Internship. This will involve industry-on-campus modules or three-day bursts where engineers, designers, and technicians teach how products are actually designed, how constraints shape engineering, how failures are analysed, and how real teams solve practical problems. This re-

moves relocation barriers, cuts bureaucracy, and injects contemporary industry thinking into classrooms instantly.

Challenge Bank: By 2030, India doesn't need more syllabus chapters. It needs thousands of real-world problem statements. A National STEM Challenge Bank – drawing challenges from ISRO, DRDO, CSIR labs, hospitals, agri-tech centres, environmental missions, and start-ups – can redefine how India

learns. Student teams solve these problems and earn micro-credits, internship priority, and national recognition. This shifts STEM from exam culture to solution culture.

Public-Private Teaching Fellowships: Across India, thousands of brilliant professionals hold expertise that rarely enters education. A Teaching Fellowship can channel this talent. Professionals teach for four hours a week for one semester, supported by contribution credits and CSR-linked incentives. They bring authenticity, current industry insight, real career visibility, and role modelling. Teachers receive reinforcement. Students gain clarity. Industry helps shape its own future workforce.

District STEM Operator Academies: India's innovation bottleneck is not just scientists. It is also the shortage of skilled operators, technicians, instrument specialists, and lab managers. These academies, jointly run by institutions and industry, can train youth in equipment handling, calibration and testing, prototyping, quality systems, and repair and maintenance. Europe's technician schools and America's community-college lab-tech programmes show how vital this layer is. India urgently needs its own.

STEM Access Pass: A simple but powerful reform is a subscription that lets any student use participating labs and sandbox rooms

across institutions. This model increases lab utilisation, gives institutions predictable revenue, ensures equitable access, and breaks the public-private resource divide. It is practical, democratic, and deployable at national scale.

Rural Talent Express: India's talent is everywhere but access is not. A Rural Talent Express fleet – mobile labs, drone classrooms, coding buses, and sensor vans – can bring monthly exposure to electronics, drone mapping, coding, sample testing, and materials basics. This ensures that the next generation of inventors may emerge from any block, not only major cities.

Access → Exposure → Capability → Innovation. This is the cycle that built the EU's maker ecosystems and America's innovation clusters. India already has the talent; it needs to give that talent early trust and early tools. If we build this Knowledge Grid now, by 2030 India will see every Class 10 student confident with real instruments, rural districts contributing innovation, strong industry-academia collaboration, a large, skilled technical workforce, and STEM becoming a national capability, not a specialised niche.

India does not require decades to transform its scientific destiny but intent, coordination, and a redesigned ecosystem.

The writer is the COO and National Coordinator of I-STEM

Too many choices?

Uncertain about your career options? Low on self-confidence? This column may help



OFF THE EDGE
Nandini Raman

I have a UG and PG degree in English Language and Literature and experience in digital marketing, graphic design, YouTube content writing and editing. I have low confidence and constantly doubt my abilities. I cleared NET and qualified for the JRF. But I am told the stipend is often delayed. I am now preparing for the state PSC exams. I don't know what else to do. Arya

Dear Arya

You have achieved things and are probably dealing with a creative burnout. Talk to a career counsellor to get some direction on what you want to do and what you are passionate about.

Public Sector Undertakings (PSUs) use the UGC-NET score to hire for roles like corporate communications or public relations. Keep an eye on the Careers page of Maharatna and Navratna PSU websites, as they release notifications specifically for NET-qualified candidates.

You could also explore UX Writing or Technical Writing that will focus on the 'micro-copy' inside apps (the buttons, error messages, and menus) or high-level documentation.

While the JRF stipend is sometimes delayed, it is a guaranteed financial cushion. Enrol for a Ph.D., at a reputable university and use the stipend (it arrives in chunks) to fund your PSC coaching. It will remove the gap year from your resume while you crack the exam and showcase you as a researcher.

I have done a B.Sc. Physics, MCA and a foundation AI/ML course. I am interested in AI and

data-related work, but not confident in coding-intensive roles. What should I focus on for an entry-level IT job? Nidhi

Dear Nidhi,

Your best entry points will be roles such as a data Analyst or BI analyst. Focus on SQL (to pull data), Excel (Advanced), and a visualisation tool like Power BI or Tableau. To be an AI data specialist or data annotator, you need an understanding of data privacy (GDPR), statistics, and tools like Informatica or Alation.

An AI quality assurance engineer runs test cases against AI outputs to ensure they are accurate, safe, and ethical. Skills required are Prompt Engineering, manual testing methodologies, and basic knowledge of Model Evaluation metrics (Precision, Recall, F1-score).

The 2026 Survival Skill Toolkit is Databases. SQL is non-negotiable. Learn Joins, Subqueries, and Window Functions; tools such as Google Vertex AI or AWS SageMaker Canvas. Focus on becoming a Pro in SQL and Power BI. These two alone can land you a job.

I completed Class 12 in 2021. I completed the first stage of the Company Secretary course but failed twice in the second. I did a course in ADVFX Plus but the salary was not great. I began working in customer service (WFH) for an e-commerce company. But the nine-hour night shift left me exhausted. I am preparing for the CDS IMA but my Maths and English are not good. I thought of SSC CPO but am not sure if I can handle the different situations of

the job. I don't know what I can do. Dev

Dear Dev,

You seem to have been through too many transitions. You need to find a path that and plays to your strengths. Meet a career counsellor and do a personality and career profiling to understand and shortlist your interests and career traits.

Consider Motion Graphics for brands. Many companies need someone to make 'explainer videos', Instagram ads, and moving logos. This pays better and usually has standard 9-to-5 hours.

Re-evaluate the CDS IMA path. What about OTA? Can you work on your English? For OTA, you only need to appear for English and General Knowledge.

You finished the first stage of CS, which means you have some knowledge of Corporate Law and Governance. What about a Legal Process Outsourcing (LPO) or Compliance Associate role? It is higher than an entry-level customer service role and has clear career growth into Risk Management.

I am a 2025 Engineering graduate working in an IT company, but want to switch to finance. Should I study for the CFA or take the CAT/GMAT and do an MBA? How can I build a strong profile as an engineer in finance? Sai

Dear Sai,

The choice between CFA and MBA depends on the Financial role you want. A Chartered Financial Analyst (CFA) is great for equity research, asset management, or portfolio management. While an MBA

in Finance is best for investment banking, private equity and so on. CFA will need self-studying now and may take upto four years across the three levels compared to a full-time two-year MBA. But the CFA costs are relatively low compared to a top B-School. While the CFA network is a global community networking may be limited compared to an MBA with campus placements and an accessible alumni base.

Consider starting your CFA Level 1 now and aim for the CAT/GMAT in 2027-2028 after gaining some IT work experience. First, bridge the gaps. Learn Financial Modelling, build three-statement models, Discounted Cash Flow, and Leveraged Buyout models. Master Python for Finance (libraries such as Panda or NumPy) and SQL. High-frequency trading and Quant funds hire engineers who can code. Apart from CFA, consider shorter certifications like FLIP or Coursera's Financial Markets (Yale) to get the basics down quickly.

Open a brokerage account and write investment reports for stocks that you own. If you can show an interviewer a report on why you bought a specific stock, you beat 90% of other applicants. Many B-schools and platforms like Unstop host finance case study competitions. Winning one is a huge resume builder.

Try to switch to a fintech project in your current role. Work on a trading platform or a banking backend. Explore moving from a general IT service firm to a company where you interact with finance professionals.

Disclaimer: This column is merely a guiding voice and provides advice and suggestions on education and careers.

The writer is a practising counsellor and a trainer. Send your questions to eduplus.thehindu@gmail.com with the subject line Off the Edge.

Time to recalibrate

What the delay of NExT means for MBBS students

David K. Pillai

When the National Medical Commission (NMC) announced that the National Exit Test (NExT) would be postponed, it reshaped preparation timelines, changed academic strategies, and disrupted the mental roadmap many students had carefully built. NExT, originally designed to replace NEET PG and the final-year MBBS exam, was meant to serve as a uniform licensure and postgraduate admission test.

With the new update that several rounds of mock exams will be held over the next three to four years before full implementation, students now have an unexpected window of time to pause, re-evaluate, and plan for the future with greater clarity. Let's look at how this decision affects students across batches.

First-year students: Their immediate priorities – building strong conceptual foundations in anatomy, physiology, and biochemistry – remain unchanged. With no pressure of an immediate format shift, they can focus on absorbing fundamentals thoroughly, which will ultimately benefit them re-



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gardless of the examination model.

Second- and third-year students: They face the most complex challenge. They must now prepare with two possibilities in mind: university exams based on traditional patterns or a shift toward clinical reasoning, which will eventually define the NExT model. This means adjusting study strategies, incorporating more case-based understanding, and staying updated on changing guidelines. It requires careful balance and time management.

Final-year students and interns: For this group, the postponement brings both relief and disruption. Many had reworked their revision cycles, internship schedules, and PG preparation plans. Now they must realign their timelines once again,

especially with NEET PG still defining postgraduate entry for the foreseeable future. Their biggest requirement right now is clarity on timelines and a predictable roadmap.

Why multiple mocks

The NMC's decision to conduct mock NExT exams over three to four years is intended to prevent a rushed rollout. This allows students to familiarise themselves with the exam structure and gradual improvement in question patterns and clinical reasoning evaluation; institutions to identify gaps in teaching and adjust curricula; and policymakers to refine scoring methods based on real data. Mock exams will also help standardise assessment across medical colleges and highlight disparities in clinical exposure and teaching

quality. Over time, this could lead to stronger integrated teaching, better case discussions, and more consistent practical training.

Student support

However, students may feel pressure to perform well even when the results are not officially binding. Managing stress and pacing preparation will be crucial in this phase. Medical colleges have to guide students through this period of uncertainty. Support mechanisms such as bridge courses focusing on applied clinical reasoning; peer-learning groups for collaborative study; counselling services to manage exam-related anxiety; and dedicated preparation pathways aligned with evolving patterns can make a difference.

Over the next few months, students can expect clearer communication from the NMC on aspects such as scoring conversion methods and weightage, the exact nature of the mock exams, timelines for different batches and how the rollouts will happen. The postponement is not a setback, it is a chance to strengthen academic grounding, improve teaching systems, and transition toward an exam model that evaluates competence rather than memorisation.

The writer is the CEO and Founding Chairman of Transworld Educare

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